

ALGEBRA

github.com/danimalabares/stack

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1. RINGS

So elementary that this is not in Stacks Project.

Theorem 1.1. *Let R be a ring and $I \subset R$ an ideal. There is a correspondence between ideals of R containing I and ideals of R/I .*

Proof. We show that there are two maps between the set of ideals of R containing I and the ideals of R/I whose compositions are the identities in the corresponding set. One map is the projection to the quotient, and the other map is taking the elements whose equivalence class lies in a given ideal.

Suppose that $\tilde{J}$ is an ideal of R/I . Define $J = \{j \in R : j + I \in \tilde{J}\}$, the set of elements that project to $\tilde{J}$. Then $I \subset J$, because $i + I = I$ is the zero of R/I which is contained in any ideal of R/I . Also notice that J is an ideal of R since for any $r \in R$, $rj + I \in \tilde{J}$ by $\tilde{J}$ being an ideal. Notice that projecting J back to R/I gives $\tilde{J}$ by definition.

Conversely, we project to the quotient. Let J be any ideal of R containing I . Let $\tilde{J}$ be the projection to R/I . Then $\tilde{J}$ is an ideal of R/I by J being an ideal of R . Consider the set J' of all elements whose projection lies in $\tilde{J}$. Then $J' = J$ by definition. $\square$

This is the essential tool in proving that R/I is a field when I is a maximal ideal

Lemma 1.2. *Let R be a ring and I a maximal ideal of R . Then R/I is a field.*

Proof. It's easy to prove that a ring is a field if and only if its only ideals are $\{0\}$ and R . By Theorem ??, the ideals of R/I are in correspondence with ideals of R containing I , but only R is such. $\square$

2. FIELDS

Just for the record, any simple field extension, i.e. the smallest field in F containing a single element, is isomorphic to either the “rational function field $k(t)/k$ ” or to one of the field extensions $k[t]/(P)$ with $P \in k[t]$ irreducible. (Stacks Project tag 09G1.)

3. MODULES

It looks like most books (at least [?] and [?]) define a module to be an abelian group along with a certain operation with elements of the ring.

Which secretly just says that

Definition 3.1. Let R be a ring. An (*left*) R -module M is an abelian group such that there exists a (left) ring morphism

$$R \rightarrow \text{End}(M).$$

Indeed, the three usual requirements correspond to:

- (1) The endomorphism corresponding to $a \in R$ respects the group structure of the group M :

$$a(x + y) = ax + ay,$$

- (2) The representation map is a map of rings

$$(a + b)x = ax + bx, \quad (ab)x = a(bx).$$

4. TENSOR PRODUCT

Definition 4.1.

Theorem 4.2 (Universal property of tensor product). *Let M_1, M_2, M be modules over a ring R . For any bilinear map $B : M_1 \times M_2 \rightarrow M$ there exists a unique homomorphism $b : M_1 \otimes M_2 \rightarrow M$ making the following diagram commute*

$$\begin{array}{ccc} M_1 \times M_2 & \xrightarrow{\pi} & M_1 \otimes M_2 \\ & \searrow B & \downarrow b \\ & & M \end{array}$$

where $\pi(m_1, m_2) = m_1 \otimes m_2$.

Remark 4.3. For $R = k$ the above property shows that $\text{Bil}(M_1 \times M_2, k) = (M_1 \otimes M_2)^*$. If we assume that the vector spaces are finite dimensional, we can take duals to obtain

$$M_1 \otimes M_2 = (M_1 \otimes M_2)^{**} = \text{Bil}(M_1 \times M_2, k)^*,$$

which is how Whitney first defined tensor products.

Here’s Misha’s trick on how to show that the universal property of tensor product implies its uniqueness. Consider the category $\mathcal{C}$ of pairs of an R -module M , and a bilinear map $M_1 \times M_2 \rightarrow M$. Then we prove that the tensor product $M_1 \otimes M_2$ is an initial object in such category, and it is easy to show that initial objects are unique.

Remark 4.4. Misha's alternative way of stating the universal property of tensor product is that there is a bijective correspondence

$$\mathrm{Hom}_R(M_1 \otimes_R M_2, M) \xleftrightarrow{\sim} \mathrm{Bil}(M_1 \times M_2, M).$$

The key observation is that the LHS are *linear* maps from a module to M while the RHS are *bilinear* maps.

Exercise 4.5 (Currying).

$$\mathrm{Hom}(M_1 \otimes_R M_2, M) = \mathrm{Hom}(M_1, \mathrm{Hom}(M_2, M))$$

Proof. This is implied by the universal property since the RHS is just $\mathrm{Bil}(M_1 \times M_2, M)$. $\square$

Exercise 4.6. Let

$$M_1 \xrightarrow{f_1} M_2 \xrightarrow{f_2} M_3 \longrightarrow 0$$

be exact. Show that

$$0 \longrightarrow \mathrm{Hom}(M_3, N) \xrightarrow{- \circ f_2} \mathrm{Hom}(M_2, N) \xrightarrow{- \circ f_1} \mathrm{Hom}(M_1, N)$$

is exact.

Proof. Injectivity of the first arrow is easy. For the middle term pick $f \in \mathrm{Hom}_R(M_2, N)$ in the kernel of $- \circ f_1$. To show that it is in the image of $- \circ f_2$ we look for $g \in \mathrm{Hom}_R(M_3, N)$ such that $f = g \circ f_2$.

Since $M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ is exact, we have an isomorphism $\bar{f}_2 : M_2/\mathrm{img} f_1 \xrightarrow{\sim} M_3$. Since $f \circ f_1 = 0$, we see that f induces a map $\bar{f}$ on the quotient since indeed, if $m_2 - m'_2 \in \mathrm{img} f_1$, we have $0 = f(m_2 - m'_2) = f(m_2) - f(m'_2)$.

Then for any $m_3 \in M_3$ we have a unique class $m_2 + M_1 \in M_2/\mathrm{img} f_1$. Thus we can define $g = \bar{f} \circ \bar{f}_2^{-1}$.

Note that this choice satisfies $f = g \circ f_2$. Indeed,

$$g \circ f_2(m_2) = \bar{f} \circ \bar{f}_2^{-1}(m_3) = \bar{f} \circ (m_2 + M) = f(m_2).$$

$\square$

Exercise 4.7. Let $R = \mathbb{Z}$. Find an example of an injective homomorphism $M_1 \rightarrow M_2$ such that $\mathrm{Hom}(M_2, N) \rightarrow \mathrm{Hom}(M_1, N)$ is not surjective.

Proof. Consider

$$\begin{array}{ccccc} 0 & \longrightarrow & 2\mathbb{Z} & \xrightarrow{i} & \mathbb{Z} \\ & & \downarrow \varphi & \nearrow \bar{\varphi} & \\ & & \mathbb{Z}/2\mathbb{Z} & & \end{array}$$

where

$$\begin{aligned} \varphi : 2\mathbb{Z} &\longrightarrow \mathbb{Z}/2\mathbb{Z} \\ 2n &\longmapsto [n]. \end{aligned}$$

Then there is no extension $\bar{\varphi} : \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ making $\varphi = \bar{\varphi} \circ i$. Indeed, note that $\varphi(2) = [1]$, while $\bar{\varphi}(2 \cdot 1) = 2\bar{\varphi}(1) = [0]$. $\square$

The idea is that any element in the image must be divisible by any integer. This is why $\mathbb{Q}$ is an injective module according to the following definition. (A precise proof of why $\mathbb{Q}$ is injective can be found at 01D6.)

Definition 4.8. An R -module is *injective* if the functor $\text{Hom}(-, M)$ is exact. That is, if the dotted arrow in the following diagram always exists:

$$\begin{array}{ccccc} 0 & \longrightarrow & M_1 & \xhookrightarrow{\quad} & M_2 \\ & & \downarrow & \nearrow \exists & \\ & & N & & \end{array}$$

Exercise 4.9. Let

$$0 \longrightarrow M_1 \xrightarrow{f_1} M_2 \xrightarrow{f_2} M_3 \longrightarrow 0$$

be an exact sequence of R -modules. Prove that

$$0 \longrightarrow \text{Hom}_R(N, M_1) \xrightarrow{f_1 \circ -} \text{Hom}_R(N, M_2) \xrightarrow{f_2 \circ -} \text{Hom}_R(N, M_3)$$

is exact

Proof. Injectivity is simple. For the middle term let $f : N \rightarrow M_2$ be in the kernel of $f_2 \circ -$, that is, $f(N)$ is killed by f_2 .

We want to define a map $g : N \rightarrow M_1$ such that $f_1 \circ g = f$. But observe that the image of f already lies in $M_1 = \text{Ker } f_2$. Thus f is already a map $f : N \rightarrow M_1$, save a more precise notation. $\square$

Exercise 4.10. Let $R = \mathbb{Z}$. Find an epimorphism (surjective homomorphism) of R -modules $M_2 \rightarrow M_3$ such that the corresponding map

$$\text{Hom}_R(N, M_2) \rightarrow \text{Hom}_R(N, M_3)$$

is not surjective for some R -module N .

Proof. Now consider

$$\begin{array}{ccccc} & & \mathbb{Z} & & \\ & \nearrow \exists & \downarrow \pi & & \\ 2\mathbb{Z} & \xrightarrow{\varphi} & \mathbb{Z}/2\mathbb{Z} & \longrightarrow & 0 \end{array}$$

for the same φ as in Exercise 4.7. Then $\pi(1) = [1]$, while any extension mapping $1 \rightarrow 2n$ gives $\varphi(2n) = 2\varphi(n) = [0]$. $\square$

Definition 4.11. An R -module N is *projective* if the functor $\text{Hom}_R(N, -)$ is exact. That is, if the dotted arrow in the following diagram always exists:

$$\begin{array}{ccccc} & & N & & \\ & \nearrow \exists & \downarrow & & \\ M_2 & \twoheadrightarrow & M_3 & \longrightarrow & 0 \end{array}$$

Exercise 4.12. A finitely generated R -module N is projective if and only if it is a direct summand of a free R -module R^n , i.e. there is a direct sum decomposition $R^n = N \oplus N'$.

Proof. Consider $R^n \simeq \bigoplus_{i=1}^n a_i R$ where N is generated by $a_1, \dots, a_n$. Apply projectivity to obtain

$$\begin{array}{ccc} & & N \\ & \swarrow \exists \psi & \downarrow \text{id} \\ R^n & \xrightarrow{\varphi} & N \longrightarrow 0 \end{array}$$

where

$$\begin{aligned} \varphi : R^n &\longrightarrow N \\ (r_1 a_1, \dots, r_n a_n) &\longmapsto \sum r_i a_i. \end{aligned}$$

This is a section of φ in the exact sequence

$$0 \longrightarrow \text{Ker } \varphi \longrightarrow R^n \begin{array}{c} \xrightarrow{\varphi} \\ \xleftarrow{\psi} \end{array} N \longrightarrow 0$$

which gives decomposition

$$R^n \simeq \text{Ker } \varphi \oplus \text{Im } \psi.$$

More explicitly, the intersection of those modules is clearly trivial while any element of R^n may be written as

$$x = \underbrace{(x - \psi\varphi(x))}_{\in \text{Ker } \varphi} + \underbrace{\psi\varphi(x)}_{\in \text{Im } \psi}.$$

For the converse we start from the diagram given in the definition of projective module, add N' to every term, arriving at $R^n \simeq N \oplus N'$. We observe that free modules are projective by constructing an extension using a basis of R^n , and then back to N by taking the action of the extension restricted to N . $\square$

In fact, currying (Exercise 4.5) is the essential ingredient for the proof that tensor product is left exact. Such result follows from two lemmas.

Lemma 4.13. *Let*

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

be an exact sequence of R -modules. Then for any R -modules N and N' , the sequence

$$0 \longrightarrow \text{Hom}_R(M_3 \otimes N', N) \longrightarrow \text{Hom}_R(M_2 \otimes N', N) \longrightarrow \text{Hom}_R(M_1 \otimes N', N)$$

is exact.

Proof. Apply $\text{Hom}_R(-, N)$, then $\text{Hom}_R(N', -)$ and then currying. $\square$

There is an (almost) converse to Exercise 4.6:

Lemma 4.14. *Let*

$$E : \quad M_1 \xrightarrow{\mu} M_2 \xrightarrow{\rho} M_3 \longrightarrow 0$$

be a complex, i.e. $\rho\mu = 0$, such that

$$0 \longrightarrow \text{Hom}_R(M_3, N) \xrightarrow{\rho_N} \text{Hom}_R(M_2, N) \xrightarrow{\mu_N} \text{Hom}_R(M_1, N)$$

is exact for any R -module N . Then E is also exact.

And this is what implies right-exactness of tensor product:

Theorem 4.15. *Let*

$$M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

be exact. Then

$$M_1 \otimes_R M \longrightarrow M_2 \otimes_R M \longrightarrow M_3 \otimes_R M \longrightarrow 0$$

is exact.

Proof. This is just Lemma 4.13 and Lemma 4.14!!! □

Exercise 4.16. Let $R = \mathbb{C}[t]$. Find an exact sequence of R -modules

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

and an R -module N such that the corresponding map

$$M_1 \otimes_R N \rightarrow M_2 \otimes_R N$$

is not injective.

Proof. Suppose the given exact sequence is just the ideal exact sequence of a point in $\mathbb{A}^1$. Then tensoring by M means thickening the point. Not having injectivity after tensoring means that tensoring the ideal by M does not give the ideal of the resulting variety. □

And from this it follows as corollary

Proposition 4.17. *Let $I \subset R$ be an ideal of a ring. Then $M \otimes_R (R/I) = M/IM$.*

Proof. Take the exact sequence $0 \rightarrow I \rightarrow R \rightarrow R/I$ and tensor by M . □

Exercise 4.18. Let I, I' be distinct maximal ideals in a ring R . Then $R/I \otimes_R R/I' = 0$.

Proof. The key observation is that since I and I' are maximal we must have $I + I' = R$. This means that $1 = f + f'$ for some $f \in I$ and $f' \in I'$. Then $1(a + I) \otimes (b + I') = (fa + I) \otimes (b + I') + (a + I) \otimes (fb + I') = 0$. □

5. ALGEBRAS

There appears to be no definition of “algebra” in Stacks Project. But there is one in [?]:

Definition 5.1. If R is a commutative ring, then a *commutative algebra* over R is a commutative ring S together with a ring morphism $R \rightarrow S$.

But actually I was expecting that S would be defined as a ring that is also an R -module. So let us note that a morphism of rings $R \rightarrow S$ gives a representation $R \rightarrow \text{End}(S)$ via left multiplication. But the other way around, given an endomorphism associated to some element in R , how do we assign an element of S so as to produce a map $R \rightarrow S$?

6. FINITELY-GENERATED AND FINITELY PRESENTED ALGEBRAS

See Stacks Project tag 00F2.

Upshot: a finitely-generated R -algebra S is such that $S \simeq R[x_1, \dots, x_n]/I$ for some ideal $I \subset R$. Another way of saying this is that there is a surjective morphism $R[x_1, \dots, x_n] \rightarrow S$. Finitely-presentedness is when $I = (s_1, \dots, s_k)$.

As an aside, recall that finitely presented k -algebras with no nilpotents are in correspondence with affine algebraic varieties.

Definition 6.1. Let $R \rightarrow S$ be a ring map.

- (1) We say $R \rightarrow S$ is of *finite type*, or that S is a *finite type R -algebra* if there exist an $n \in \mathbf{N}$ and a surjection of R -algebras $R[x_1, \dots, x_n] \rightarrow S$.
- (2) We say $R \rightarrow S$ is of *finite presentation* if there exist integers $n, m \in \mathbf{N}$ and polynomials $f_1, \dots, f_m \in R[x_1, \dots, x_n]$ and an isomorphism of R -algebras $R[x_1, \dots, x_n]/(f_1, \dots, f_m) \cong S$.

IA: for modules, the definition of finitely generated R -module is that there exists a surjective morphism $R^n \twoheadrightarrow S$. So: f.g. algebra means elements are polynomials on the generators with coefficients in the ring, and f.g. module means elements are linear combinations on the generators with coefficients in the ring.

7. NOETHERIAN AND ARTINIAN RINGS

Definition 7.1. A ring is called *Noetherian* if any increasing chain of ideals $I_1 \subset I_2 \subset \dots$ stabilizes. It is called *Artinian* if any descending chain of ideals $I_1 \supset I_2 \supset \dots$ stabilizes.

Exercise 7.2. A ring is Noetherian if and only if all its ideals are finitely generated.

Proof. First we prove that if not every ideal is finitely generated then R cannot be Noetherian. Indeed, if I is not finitely generated we construct an ascending chain by of ideals by starting with any generator of I and adding other generators one by one.

For the converse suppose that all ideals of R are finitely generated and pick an ascending chain $I_1 \subset I_2 \subset \dots$. Consider $I = \bigcup_i I_i$, which is an ideal since for any $a \in I$ there is i such that $a \in I_i$ so that $ra \in I_i$ for any $r \in R$. However, since it is finitely generated, all the generators have to be in some I_j , and thus the chain stabilizes. $\square$

Theorem 7.3 (Hilbert basis theorem). *If R is Noetherian then so is $R[x]$.*

Exercise 7.4.

Proof. $k[x_1, x_2, \dots]/(x_1^2, x_2^2, \dots)$, $I_n = (x_n, x_{n+1}, \dots)$ then $I_1 \supset I_2 \supset \dots$ $\square$

Exercise 7.5. If A is Noetherian then so is $A[[t]]$.

Proof. Pass from an ideal in $A[[t]]$ to A by taking the coefficients of the monomials of minimal degree of each element in the ideal. Then take a finite set of generators. Then go back to $A[[t]]$ by taking the polynomials where the generators were taken. Then... $\square$

8. LOCALIZATION

The general recipe for localization is that $\mathcal{O}_U = \mathcal{O}_X[f^{-1}]$ where $U = X \setminus V(f)$. That is, localizing gives the sheaf of the variety without the zero set.

Definition 8.1. Let R be a ring and $S \subset R$ a multiplicative subset of R (containing 1). The *localization* of R at S is the ring RS^{-1} (where S^{-1} is just a set with the same elements as S but with a -1 exponent) with the operations of $(rs^{-1})(r_1s_1^{-1}) = rr_1s^{-1}s_1^{-1}$, $rs^{-1} + r_1s_1^{-1} = (rs_1 + r_1s)(s^{-1}s_1^{-1})$ and the identification given by $rs^{-1} = r_1s_1^{-1} \iff rs_1 = r_1s$. (Check this definition if R has zero divisors!)

In a simple case, consider a ring R and $f \in R$. The *localization of R at f* is $R[t]/(ft - 1) := R[f^{-1}] = R(f)$. The quotient basically says that t is the inverse of f .

Example 8.2.

- Dyadic numbers: $\mathbb{Z}[2^{-1}]$, which are fractions of integers with denominators which are powers of 2.
- Laurent polynomials: $\mathbb{C}[t, t^{-1}]$.

Lemma 8.3. *If $R[f^{-1}] = 0$ then f is nilpotent.*

Proof. If $R[f^{-1}] = 0$ then $1 \in (ft - 1)$, which means that $1 = (1 - ft)p(t)$ for some $p(t) \in k[t]$. Then... $\square$

The localization a ring is not always a local ring. But if S is a prime ideal, then it is.

Exercise 8.4. Prove that localization of a Noetherian ring is Noetherian.

Proof. Let R be a Noetherian ring and $S \subset R$. Let $I_1 \subset I_2 \subset \dots$ be a chain of ideals in the localization $R[S^{-1}]$. Let $\pi : R \rightarrow R[S^{-1}]$ the canonical projection $r \mapsto r$.

Let I be any of the I_i . Consider $\pi^{-1}(I)$. This gives a chain of ideals in R , which must stabilize. To go back to the localization simply recall that $\pi^{-1}(I)[S^{-1}] = I$. $\square$

9. FINITE AND INTEGRAL RING EXTENSIONS

The upshot is: let $\varphi : R \rightarrow S$ be a ring map. An element $s \in S$ is integral over R if there is a monic polynomial with coefficients in R that gives zero when you put s instead of the variable.

But it might be convenient to think about finite (or, for humans, finitely-generated) R -modules: x is integral over R if and only if $R[x]$ is a finitely-generated R -module (i.e., not only finitely generated as an algebra but also as a module). See Zvi Rosen's video. The forward implication is: suppose x is integral, then $x^n + r_1x^{n-1} + \dots + r_n = 0$, so $x^n = -r_1x^{n-1} - \dots - r_n$, which "says" $R[x] \cong \bigoplus_{k=0}^{n-1} Rx^k$.

So maybe the details of that are not entirely trivial (Zvi uses some theorem, and Stacks Project does lots of things). But the point is: I think that after applying Spec, the fibers of an integral extension are **finite**. Which of course has all to do with the finite-moduleness I was trying to explain.

10. GRÖBNER THEORY

So far here's some rather unformatted notes from Lin's talk at Commutative Algebra school in Recife 2025.

Would SAGBI bases be related to this?

square free monomial ideals preserve regularity upon Grobner deformation i.e. initial ideal!

Constantini-Fouli-Goel-Lin-Lindo-Liske-Mostafazadehfard, 2025.

<https://sites.psu.edu/kul20/>

How is Lin using the Reese algebra here? Is it true that the initial ideal of the Gröbner deformation coincides with the initial ideal "of the Reese algebra"?

REFERENCES